

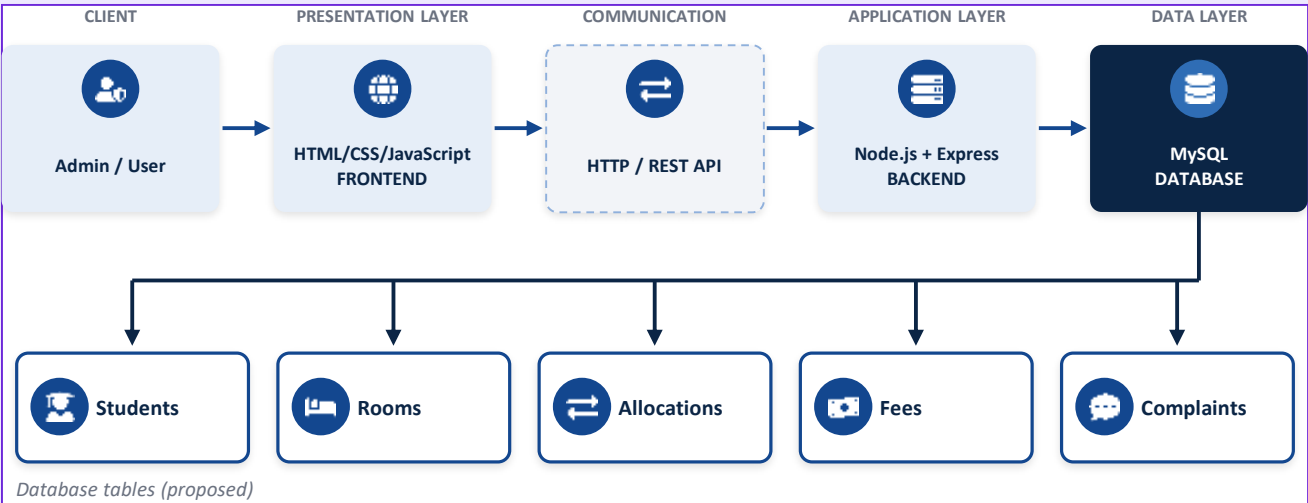


Smart Hostel Management System

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Database Systems Engineering and Distributed Backend Development (25CS1302E) | KLH Bachupally Campus, CSE | PBL 2026-27

System Architecture



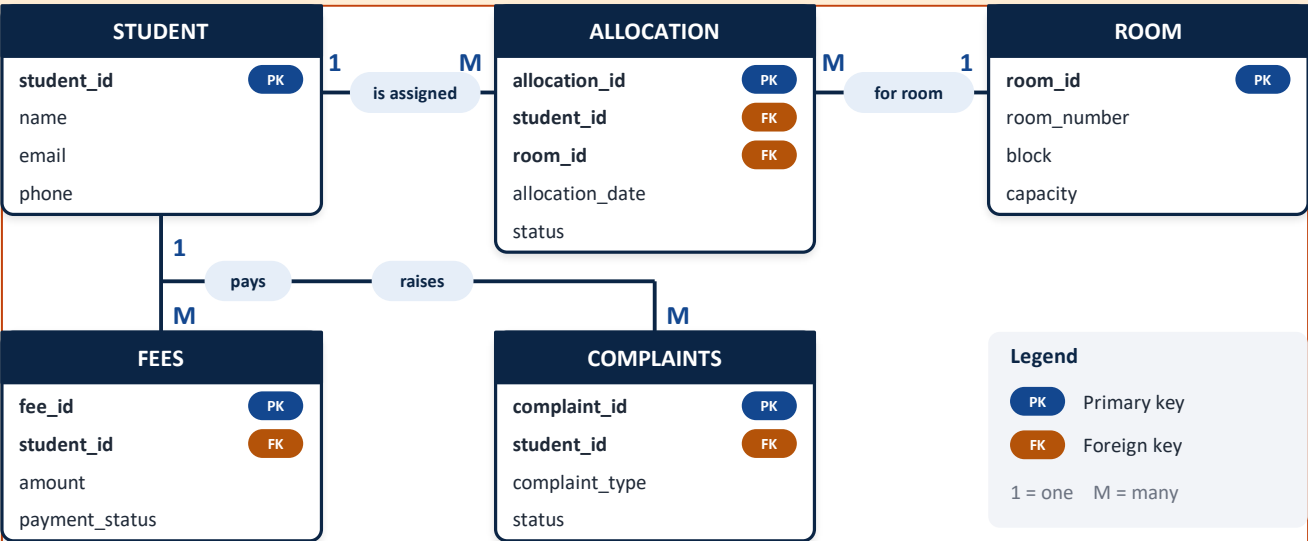
Abstract

- A centralized web application that helps hostel administrators manage students, rooms, allocations, fees and complaints, built with Node.js, Express and MySQL.
- Replaces manual record keeping with structured data and faster retrieval.

Tools & Technologies

- Front-End:** HTML, CSS, JavaScript
- Back-End:** Node.js + Express.js
- Database:** MySQL
- Tools:** VS Code, Git & GitHub

ER Diagram & Database Design



Problem Statement & Objectives

- Problem:** Manual hostel records cause duplication, errors and slow searching.
- Obj 1:** Centralize student, room, fee and complaint data in MySQL.
- Obj 2:** Provide CRUD and room allocation for authorized users.
- Obj 3:** Reduce redundancy and improve information retrieval.
- Scope:** Students, rooms, allocation, fees, complaints, admin.

Feasibility Study

- Technical:** Standard, well-documented tools.
- Economic:** Fully open-source stack, no licensing cost.
- Operational:** Simple admin interface with role-based access.
- Schedule:** 8 phases, from requirement analysis to final demo.

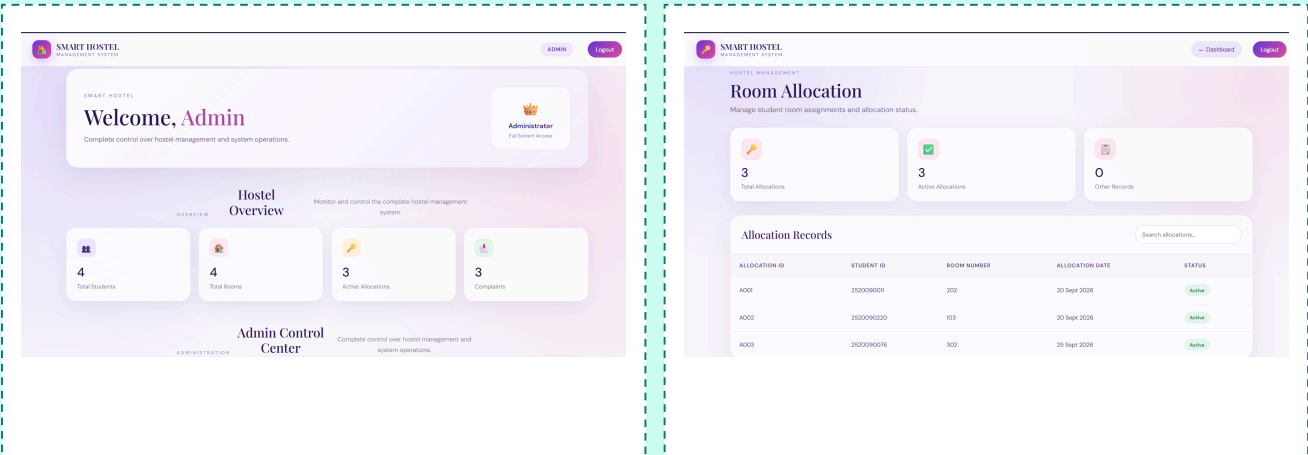
Existing vs Proposed System

Feature	Existing System	Proposed System
Records	Paper / spreadsheets	Centralized MySQL database
Room allocation	Manual, hard to track	Assign / change rooms with history
Fees	Manual tracking	Payment records and status
Complaints	Manual tracking	Registered with status updates
Limitation / Gap	Redundancy, errors, slow search	Structured data, fast retrieval, role-based access

Results & Testing

- Expected: working CRUD for all five modules.
- Results to be added after implementation and testing.

Implementation Screenshots



Conclusion

- A centralized, database-driven system to improve hostel administration.
- Aims to cut manual effort and redundancy.

Future Scope

- Mobile app, online fee payment, SMS / email alerts
- Cloud deployment, Docker, microservices

